

A Multi-Subject Retrieval-Augmented Generation System for NCERT Science Question Answering Across Physics, Chemistry, Mathematics, and Biology

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Abstract—We present a production-ready, multi-subject Retrieval-Augmented Generation (RAG) system for question answering grounded in the NCERT Class 11 and 12 Science curriculum, spanning Physics, Chemistry, Mathematics, and Biology. We construct a RAG-optimised corpus of 14,450 enriched chunks from 99 chapters across four textbooks, each annotated with content-type classification, difficulty level, extracted key terms, formulas, and *dual retrieval texts* optimised separately for BM25 keyword matching and dense embedding retrieval. A five-stage retrieval pipeline integrates BM25 sparse retrieval with BAAI/bge-base-en-v1.5 dense embeddings via Reciprocal Rank Fusion, followed by cross-encoder reranking with subject-aware ($\alpha=2.0$) and chapter-aware ($\beta=4.0$) score boosting and parent-child chunk expansion. Generation is performed by Qwen2.5-1.5B-Instruct. Evaluation on a 573-question holdout set spanning four subjects, five question types, and three difficulty levels shows that the full pipeline attains Recall@5 of 91.6% (95% CI [89.4, 93.7]), Recall@10 of 94.6% (95% CI [92.7, 96.5]), and MRR of 0.810 (95% CI [0.782, 0.835]). Three of the four pipeline stages contribute statistically significant gains by paired McNemar tests ($p < 10^{-3}$); the final subject/chapter-boost stage is robust but marginal at this evaluation scale. Counter to the conventional assumption that dense retrieval dominates BM25, on this curriculum-aligned corpus BM25 alone achieves 84.8% Recall@10 versus 79.4% for dense retrieval – yet the hybrid combination outperforms both (89.5%), confirming the complementary role of the two signals. Manual analysis of all 31 nominal failures shows only ~ 12 are genuine retrieval errors (effective Recall@10 $\approx 98\%$). Against five open-source LLM baselines spanning four families (Meta, Mistral, Alibaba, Google) and 2.6B–8B parameters, our 1.5B+RAG system achieves comparable BERTScore F1 (0.809 vs. 0.803–0.810) at 1.7–5.3 $\times$ smaller generator size, with substantially narrower confidence intervals reflecting RAG’s per-question consistency. End-to-end latency is 9.4 s on a single Colab L4 GPU at batch size 1; retrieval

contributes only 250 ms (2.7%). The fully open-source, Colab-compatible implementation provides a reproducible multi-subject baseline for curriculum-aligned educational QA research.

Index Terms—Retrieval-Augmented Generation, Educational AI, NCERT, Multi-Subject QA, FAISS, BM25, Cross-Encoder Reranking, Hybrid Retrieval, Curriculum-Aligned QA

I. INTRODUCTION

The National Council of Educational Research and Training (NCERT) textbooks define the canonical curriculum for over 250 million students in India [1]. Despite their pedagogical importance, automated question-answering (QA) systems targeting NCERT content remain underexplored – particularly for multi-subject, multi-grade coverage.

Large Language Models (LLMs) offer fluent generation but frequently hallucinate curriculum-specific facts when deployed without grounding [2]. Sharma et al. [3] demonstrated this empirically: Llama-4-Scout-17B F1 collapses from 0.894 on SQuAD to 0.395 on NCERT-QA without retrieval context. Retrieval-Augmented Generation (RAG) [5] addresses this by conditioning generation on retrieved passages, combining factual precision with generative fluency.

Existing NCERT QA systems target single subjects [3], [4] or narrow grade ranges. Multi-subject corpora introduce a unique challenge: *cross-subject contamination* – queries about “equilibrium” may retrieve Chemistry chunks (chemical equilibrium) when the target is Physics (mechanical equilibrium), and “derivative” retrieves Mathematics chunks when the context is Chemistry (functional-group derivatives).

A. Contributions

This work makes five primary contributions:

- 1) **Multi-subject RAG corpus.** A production-ready dataset of **14,450 enriched chunks** spanning 99 chapters across Physics (21 ch.), Chemistry (19 units), Mathematics (27 ch.), and Biology (32 ch.) for Classes 11–12, with seven content types, dual retrieval texts, extracted formulas and key terms, structured problem objects, and PDF-location-aware bounding boxes.
- 2) **Subject-aware hybrid retrieval pipeline.** A five-stage architecture – hybrid BM25/dense fusion via Reciprocal Rank Fusion (RRF), cross-encoder reranking, additive subject/chapter boosting, and sibling chunk expansion – designed for cross-subject disambiguation while avoiding the overhead of per-subject indices.
- 3) **Comprehensive empirical evaluation.** A 573-question holdout set covering four subjects, five question types, and three difficulty levels, with systematic five-condition ablation, paired McNemar significance tests, paired-bootstrap effect sizes, and a manually-categorised failure analysis. The full pipeline achieves 94.6% **Recall@10** (95% CI [92.7, 96.5]), **MRR** 0.810 (CI [0.782, 0.835]), and **BERTScore F1** 0.805.
- 4) **A counter-intuitive empirical finding.** On curriculum-aligned QA dominated by definitional and exercise-style queries, BM25 *outperforms* dense retrieval (84.8% vs 79.4% R@10), yet hybrid fusion beats both (89.5%) and cross-encoder reranking adds another +4.2 pp. This justifies hybrid retrieval on principled grounds rather than as a default heuristic.
- 5) **Parameter-efficiency benchmark.** Comparison against five open-source LLM baselines spanning four families (Meta, Mistral, Alibaba, Google) and 2.6B–8B parameters shows that our 1.5B+RAG system matches the BERTScore F1 of much larger zero-shot models, with substantially narrower confidence intervals reflecting RAG’s per-question consistency.

The system scales a prior single-subject Physics prototype (2,619 chunks, 84 questions) by $5.5\times$ in corpus size and $6.8\times$ in evaluation coverage.

II. RELATED WORK

A. Foundational RAG Framework (Base Paper)

The retrieval-augmented generation framework introduced by Lewis et al. [5]—which we identify as the *base paper* for this work—couples a parametric language model with a non-parametric memory of dense vector indices, retrieving top- k passages and prepending them to an autoregressive decoder. The framework was subsequently extended by REALM [6], which jointly pre-trains the retriever and the language model end-to-end, and by Fusion-in-Decoder [7], which aggregates evidence from multiple passages within the decoder rather than concatenating them at the prompt level. Our pipeline preserves the original Lewis-style retrieve-then-generate architecture but augments it with a five-stage retrieval cascade designed for

the cross-subject disambiguation challenges absent from the open-domain QA settings these foundational works target.

B. Dense and Sparse Retrieval

Dense Passage Retrieval (DPR) by Karpukhin et al. [8] demonstrated that learned dense encoders outperform classical BM25 on open-domain QA, establishing the dual-encoder paradigm that motivates our use of `bge-base-en-v1.5`. BERT [9] provides the contextual encoding backbone that this and subsequent retrievers build on; Sentence-BERT [10] adapts the architecture for efficient sentence-level embeddings, a key prerequisite for chunk-level retrieval. ColBERT [11] introduces late-interaction matching that interpolates between bi-encoder efficiency and full cross-encoder accuracy. Hybrid retrieval combining BM25 with dense embeddings is conventionally reported to outperform either alone [12], although the relative strength of the two signals is corpus-dependent—a phenomenon we observe and quantify on curriculum-aligned QA. Cross-encoder reranking by Nogueira and Cho [13] further improves context quality after a fast first-stage retrieval, motivating Stage 2 of our pipeline.

C. Educational QA and NCERT Datasets

Sharma et al. [3] introduced *PustakAI* and the NCERT-QA dataset (739 Q&A pairs, Grades 6–8), showing that curriculum-specific context is essential. Their meta one-shot strategy delivered faithfulness of 0.85 and F1 of 0.46 with Llama-4-Scout-17B. Rao et al. [4] (IISc Bangalore) built a multi-task tutoring system combining GPT-4o-mini QA (BERTScore = 0.8495 descriptive, 0.8055 numerical), fine-tuned T5 [16] question generation (ROUGE-L = 0.2815), and BAAI/bge-reranker-large in a two-stage retrieval pipeline. Earlier work in this space includes TextbookQA [22] (high-school science textbook QA over diagrams and text) and SciQ [23] (crowdsourced multiple-choice science questions). Both Sharma et al. and Rao et al. target a single subject or grade band; our work extends to four subjects across two grade levels and explicitly tackles the multi-subject disambiguation problem. The MMLU benchmark [24] provides additional context for STEM-domain LLM evaluation but does not test retrieval grounding.

D. QA Datasets and Evaluation Methodology

Reading-comprehension QA was popularised by SQuAD [19] (100K Wikipedia-grounded questions) and extended to realistic web queries by Natural Questions [20]. Multi-hop reasoning questions requiring evidence aggregation across documents were introduced by HotpotQA [21], motivating our sibling-chunk expansion strategy. Hierarchical text segmentation, foundational to our chunking pipeline, traces back to Barzilay and Elhadad [25]. Standard retrieval evaluation employs Recall@K and follows the BEIR benchmark suite [26]; for generation, ROUGE [27] captures lexical overlap while BERTScore [28] provides contextual semantic similarity. More recent reference-free evaluators include RAGAS [30] and G-Eval [29]. Our evaluation

combines retrieval metrics (Recall@K, MRR), generation metrics (ROUGE-L, BERTScore, Concept Coverage), and statistical significance testing (McNemar, paired bootstrap) across a 573-question multi-subject holdout with 95% bootstrap confidence intervals.

E. Vector Search Infrastructure and Reasoning

Our dense index is built on FAISS [31], the de-facto standard for billion-scale similarity search. For complex multi-step reasoning—particularly relevant for numerical Physics problems and Mathematics derivations in our corpus—chain-of-thought prompting [17] elicits step-by-step decomposition from instruction-tuned LLMs; we incorporate this stylistic guidance in our type-conditioned prompts (Section V). Our pipeline extends hybrid retrieval with subject-aware and chapter-aware score boosting, addressing the cross-subject contamination problem unique to multi-subject corpora. Unlike approaches that maintain per-subject indices, our additive boost avoids storage and maintenance overhead while retaining single-index retrieval latency.

III. DATASET CONSTRUCTION

A. Source Material

Four NCERT textbooks (Reprint 2025–26) serve as source material, totalling 2,409 pages across 99 chapters:

- **Physics:** 625 pages, 21 chapters (7 Class 11 per 2023+ rationalised edition, 14 Class 12).
- **Chemistry:** 591 pages, 19 units (9 Class 11, 10 Class 12).
- **Mathematics:** 713 pages, 27 chapters (14 Class 11, 13 Class 12).
- **Biology:** 480 pages, 32 chapters (19 Class 11, 13 Class 12).

Each PDF was processed through a document intelligence extraction system, producing JSON chunks with page-level bounding boxes for every text segment, figure description, and table.

B. Five-Stage Processing Pipeline

Figure 1 illustrates the construction pipeline.

Stage 1 – Parse & Merge: Per-subject JSON splits (7–8 files per book) are merged with global page-number offset correction, yielding approximately 6,500 raw chunks per subject.

Stage 2 – Structure: Chapter boundaries are detected via subject-specific heading patterns: uppercase CHAPTER X for Biology/Physics, Unit X for Chemistry, and logo-embedded chapter markers with section X.1 detection for Mathematics. Exercise, summary, and Points-to-Ponder zones are identified via regex pattern matching across text, logo, and table chunk types. Section/subsection hierarchy is extracted from numbered headings (e.g., 3.2.1). Fourteen distinct exercise detection patterns handle format variations.

Stage 3 – Validate: A 14-check diagnostic suite is applied: duplicate chunk IDs (8 fixed), null class/chapter fields (11 fixed), empty content (2 removed), corrupt equation-fragment section titles (188 cleaned), and noise chunks including page headers and reprint notices (112 removed).

Stage 4 – Chunk Optimisation: Short chunks (< 80 characters) are merged with adjacent same-chapter neighbours (2,465 merged). Oversized chunks (> 3,000 characters) are split at paragraph boundaries with parent–child relationship tracking (18 split). This reduces the corpus from 16,906 to 14,450 chunks while improving average chunk quality.

Stage 5 – RAG Enrichment: Each chunk is annotated with the fields described in Section III-D.

C. Per-Subject Breakdown

Each subject presents distinct processing challenges:

a) *Physics (3,319 chunks):* The 2023+ rationalised Class 11 edition reduced content from 14 to 7 chapters, leaving gaps around Thermodynamics, Oscillations, and Waves. Bounding boxes are essential because many chapters intersperse diagrams with derivations.

b) *Chemistry (4,250 chunks):* Chemistry uses Unit X rather than Chapter X; our parser handles both forms. The dataset has the highest diagram density (643 diagram chunks; 15.1% of the subject corpus) due to molecular-structure illustrations.

c) *Mathematics (4,584 chunks):* Mathematics has by far the most exercise chunks (1,352, or 29.5% of the subject corpus). Section-level exercises require sub-chapter detection: the parser tracks X.Y patterns to associate each exercise with its parent section.

d) *Biology (2,297 chunks):* Biology has zero solved examples because the textbook uses inline explanations rather than numbered worked examples. Concept-type chunks therefore receive richer key-term metadata.

D. Chunk Schema

Figure 2 illustrates the enriched chunk schema. Each record contains seven nested objects:

- **content_type:** One of seven categories – concept, example, exercise, diagram, table, summary, points_to_ponder.
- **searchable_text:** BM25-optimised text comprising chapter prefix, section title, raw content, and appended key terms and formulas.
- **embedding_text:** Dense-retrieval-optimised text with structured context prefix: [Subject ChN Title | Section] content.
- **key_terms:** Extracted from bold and italic markers, up to 15 per chunk.
- **formulas:** Extracted via LaTeX patterns and common equation patterns, up to 10 per chunk.
- **difficulty:** Heuristic assignment based on section position and content type.
- **problems:** Structured object with `is_problem`, `problem_type`, `question_text`, and `solution_text`.
- **location:** Source PDF, page index/number, and normalised bounding-box coordinates $\{l, t, r, b\} \in [0, 1]^4$.

Five-stage corpus construction pipeline

From 4 raw NCERT PDFs (2,409 pp) to a validated 14,450-chunk corpus

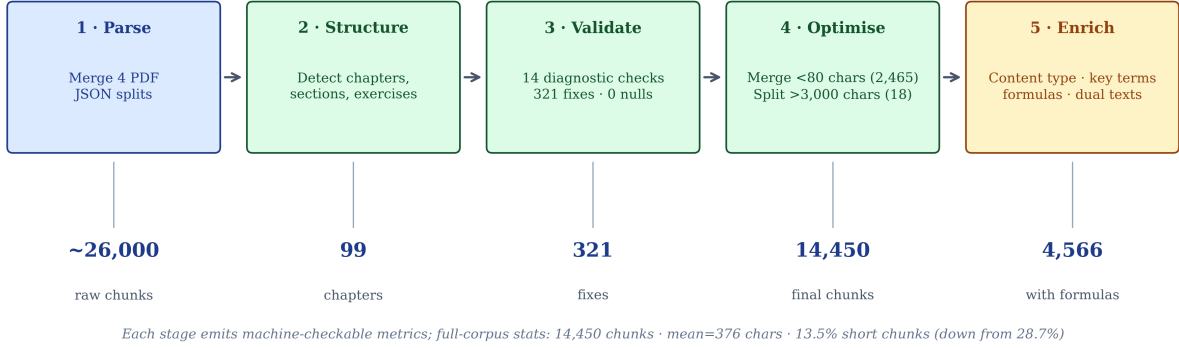


Fig. 1. Five-stage dataset construction pipeline from raw NCERT PDFs to RAG-optimised chunks. Bottom row shows final corpus statistics after diagnostic validation, chunk optimisation, and RAG enrichment.



Fig. 2. Enriched chunk schema. Each record contains seven nested objects covering hierarchy, content, problems, visuals, location, metadata, and RAG-optimised retrieval texts.

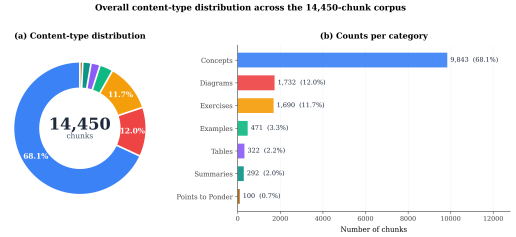


Fig. 4. Overall content-type distribution across the 14,450-chunk corpus.

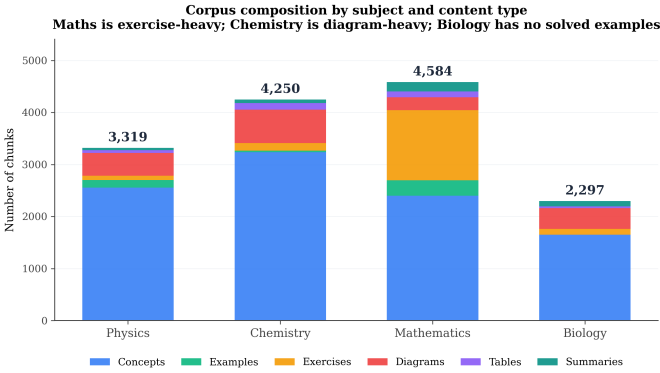


Fig. 3. Corpus composition by subject and content type. Mathematics has the most exercise chunks (1,352); Chemistry has the most diagram descriptions (643); Biology has no solved examples.

E. Corpus Statistics

Figure 3, Figure 4, and Table I present the corpus breakdown. Content-length statistics after splitting: min = 1, max = 3,194, mean = 376, median = 247 characters. Short chunks (< 100 characters) comprise 13.5% of the corpus, reduced from 28.7% pre-merging.

IV. SYSTEM ARCHITECTURE

Figure 5 presents the end-to-end architecture comprising five retrieval stages and prompted generation.

A. Embedding and Indexing

Dense embeddings are computed using BAAI/bge-base-en-v1.5 [18] (768-dimensional, instruction-tuned) over the embedding_text field. Vectors are stored in a FAISS [31] IndexFlatIP index for exact inner-product search; because embeddings are L2-normalised, this is equivalent to cosine similarity. A parallel BM25 [14] index (rank_bm25) operates on searchable_text. Together the two indices consume ≈ 44 MB of RAM.

B. Stage 1: Hybrid Retrieval via Reciprocal Rank Fusion

For each query q , both BM25 and dense retrieval return the top- N candidates ($N = 100$). BM25 scoring is:

$$s_{\text{BM25}}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d) (k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \quad (1)$$

where $f(t, d)$ is the term frequency, $k_1 = 1.5$, $b = 0.75$. Dense scoring is:

$$s_{\text{dense}}(q, d) = \cos(\mathbf{E}_q, \mathbf{E}_d) = \frac{\mathbf{E}_q \cdot \mathbf{E}_d}{\|\mathbf{E}_q\| \|\mathbf{E}_d\|} \quad (2)$$

TABLE I
CORPUS STATISTICS BY SUBJECT

Subject	Chunks	Concepts	Examples	Diagrams	Exercises	Tables	Summaries
Physics	3,319	2,554	147	441	82	53	42
Chemistry	4,250	3,239	29	643	145	124	70
Mathematics	4,584	2,400	295	248	1,352	110	179
Biology	2,297	1,650	0	400	111	35	101
Total	14,450	9,843	471	1,732	1,690	322	392

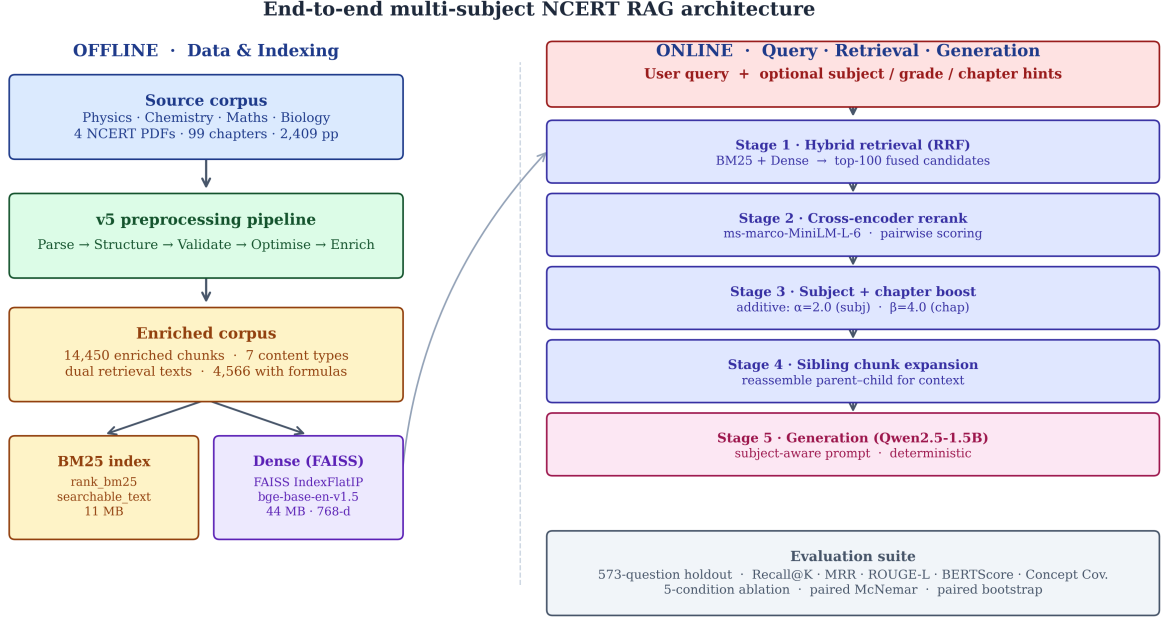


Fig. 5. End-to-end multi-subject RAG pipeline. *Left*: Offline path builds BM25 and FAISS indices from 14,450 enriched chunks. *Right*: Online path performs five-stage retrieval culminating in subject-aware prompted generation.

Scores are fused via Reciprocal Rank Fusion [15]:

$$s_{\text{RRF}}(d) = \sum_{r \in \{\text{BM25}, \text{Dense}\}} \frac{1}{\text{rank}_r(d) + k} \quad (3)$$

where $k = 60$. RRF eliminates the need for score normalisation across heterogeneous retrieval systems.

C. Stage 2: Cross-Encoder Reranking

The top-100 RRF candidates are reranked by a cross-encoder [13] (cross-encoder/ms-marco-MiniLM-L-6-v2):

$$s_{\text{CE}}(q, d) = \text{CrossEncoder}([q; d]) \quad (4)$$

where $[q; d]$ denotes concatenation. The 22M-parameter MiniLM produces a relevance logit per pair with ≈ 1.4 ms per pair on a Colab L4 GPU.

D. Stage 3: Subject and Chapter Boosting

Subject and chapter metadata are incorporated as additive score boosts:

$$s'(d) = s_{\text{CE}}(d) + \alpha \cdot \mathbb{I}[\text{subj}(d) = \text{subj}(q)] + \beta \cdot \mathbb{I}[\text{ch}(d) = \text{ch}(q) \wedge \text{subj}(d) = \text{subj}(q)] \quad (5)$$

with $\alpha = 2.0$ (subject boost) and $\beta = 4.0$ (chapter boost). This lightweight mechanism avoids per-subject indices while resolving ambiguous terminology. As shown empirically in Section IX-A, the system is robust to these constants over a wide range; the chosen values lie on the saturation plateau of the MRR curve.

E. Stage 4: Sibling Chunk Expansion

For chunks split during dataset Stage 4, all siblings sharing the same `parent_chunk_id` are reassembled in the generation context:

$$\text{Context}(c_i) = \bigoplus_{j=1}^{|S_p|} c_j \quad \text{where} \quad S_p = \{c : \text{parent}(c) = p\} \quad (6)$$

TABLE II
COMPONENT SELECTION AND RATIONALE

Component	Choice	Rationale
Embeddings	bge-base-en-v1.5	BEIR SOTA, 768-dim
Vector Store	FAISS IndexFlatIP	Exact cosine, < 10 ms
Sparse Index	BM25 (rank-bm25)	Keyword precision
Score Fusion	RRF ($k=60$)	No normalisation needed
Reranker	ms-marco-MiniLM-L-6-v2	Fast cross-encoder
LLM	Qwen2.5-1.5B-Instruct	Instruction-tuned
Subject Boost	$\alpha=2.0$	On saturation plateau
Chapter Boost	$\beta=4.0$	MRR saturates at $\beta \geq 4$

This ensures the LLM receives complete paragraphs. Expansion is applied only at the generation-context stage; retrieval metrics are reported on the pre-expansion flat ranking.

F. Stage 5: Prompted Generation

Generation uses Qwen2.5-1.5B-Instruct with subject-specific prompts:

```
System: You are an expert NCERT {Subject} tutor for
        Class {Grade}. Topic: {Chapter}. Answer
        using
        ONLY the provided sources. {Style}
User:   SOURCES: {context}
        QUESTION: {query}
```

where *Style* adapts by question type: step-by-step derivation for numerical, direct statement for definitions. Token budgets range from 300 (factual/definition) to 420 (numerical).

G. Component Selection

Table II summarises component choices.

V. METHODOLOGY

A. End-to-End Query Flow

For each evaluation question q with optional metadata ($\text{subj}_q, \text{ch}_q, \text{type}_q$):

- 1) **Query augmentation.** BM25 input is prefixed with lightweight hints: `{subject} Chapter {N} ch{N}, plus ``formula calculation numerical``` when $\text{type}_q=\text{numerical}$. Dense input remains the raw query to preserve semantic fidelity.
- 2) **Hybrid retrieval.** Compute Equations 1–3; retain top-100 fused candidates.
- 3) **Cross-encoder reranking.** Rerank with Equation 4.
- 4) **Subject/chapter boosting.** Apply Equation 5; select top-15 flat IDs.
- 5) **Sibling expansion** (Equation 6). Reassemble parent–child groups for the generation context.
- 6) **Context assembly.** Deduplicate and truncate expanded chunks to a 5,000-character budget, interleaving “[Source N]” markers.
- 7) **Prompted generation.** Feed the chat-templated prompt to Qwen2.5-1.5B with a type-conditioned token budget.

B. Preprocessing and Tokenization

The BM25 tokenizer lower-cases, strips punctuation, and removes tokens shorter than 3 characters or in the English stopword list, with a curated *keep list* (“force”, “velocity”, “mitochondria”, “chapter”, ...) that protects domain-relevant terms. The dense encoder receives raw query text.

C. Generation Pipeline

Generation is deterministic (`do_sample=False`) so all reported numbers are reproducible. Prompts are type-conditioned: numerical questions receive an explicit “show the formula, substitute values step-by-step” directive; definition/factual prompts request “a clear, direct answer.”

VI. EVALUATION METHODOLOGY

A. Evaluation Set Construction

The 573-question evaluation set (Figure 7) is constructed from two sources:

NCERT exercises (379 questions, 66.1%): End-of-chapter exercise questions extracted from the textbooks, linked to source chunk IDs via the `exercises_index`.

Key-term-generated questions (194 questions, 33.9%): Definitional questions generated from extracted bold/italic key terms: “What is {term}? Explain in the context of {chapter}.” We discuss this construction’s failure modes in Section VII-G.

Each question carries subject, grade, chapter meta-data, question type, difficulty, reference chunk ID(s), a `recall_any_of` list for lenient matching, and a reference answer. Five question types are represented: definition (226), conceptual (200), numerical (78), factual (46), and reasoning (23).

B. Metrics

Retrieval metrics.

- **Recall@K** ($K \in \{5, 10\}$): Lenient evaluation – a hit if any chunk in the `recall_any_of` list appears in the top- K results.
- **MRR**: $\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}$.

Generation metrics.

- **ROUGE-L** [27]: Longest common subsequence F-measure.
- **BERTScore F1** [28]: Contextual embedding-based semantic similarity.
- **Concept Coverage**: $\text{CC}(a, C) = |\{c \in C : c \in a\}|/|C|$, where a is the generated answer and C is the set of reference key concepts.

All metrics include 95% bootstrap confidence intervals ($n = 1,000$).

C. Ablation Conditions

Five conditions isolate each component’s contribution:

- 1) **BM25-only:** Pure keyword retrieval on `searchable_text`.
- 2) **Dense-only:** Pure FAISS cosine on `embedding_text`.
- 3) **Hybrid RRF:** BM25 + Dense with RRF, no reranking.
- 4) **Hybrid + Rerank:** RRF + cross-encoder, no boosting.

Mathematical formulation of the five retrieval stages

Each stage's score function is shown; arrows indicate signal flow

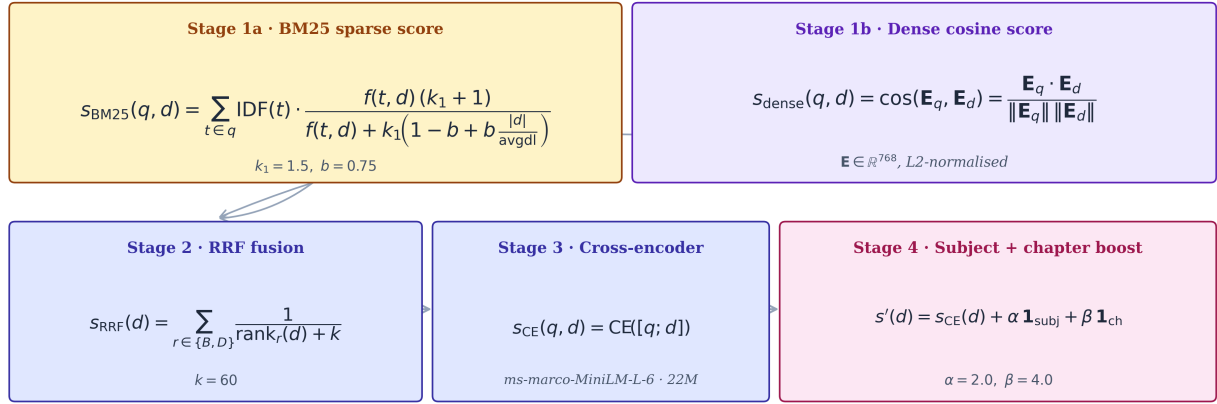


Fig. 6. Mathematical formulation of the five retrieval stages showing BM25 scoring, dense cosine similarity, RRF fusion, cross-encoder reranking, and subject/chapter boosting.

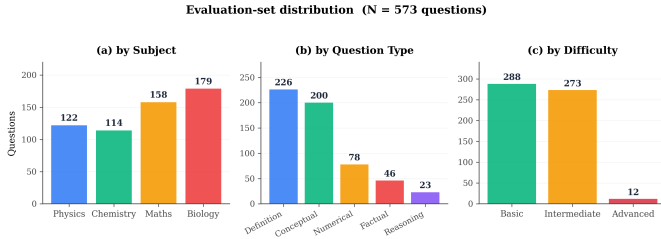


Fig. 7. Evaluation set distribution across (a) subjects, (b) question types, and (c) difficulty levels. Total: 573 questions.

5) **Full pipeline:** RRF + rerank + boost + sibling expansion. A sixth comparison condition (**Zero-shot LLM:** Qwen2.5-1.5B without retrieval, scored on a matched 200-question subset) is reported alongside.

VII. RESULTS AND ANALYSIS

A. Ablation Study

Table III and Figure 8 present the main ablation. The largest single gain comes from RRF fusion over Dense-only (+10.1 pp at Recall@10); cross-encoder reranking adds another +4.2 pp; subject/chapter boosting adds a final +0.9 pp. The full pipeline reaches 94.6% R@10 with a 95% CI of [92.7, 96.5].

A counter-intuitive finding stands out. **BM25 alone outperforms dense retrieval** on this corpus (84.8% vs 79.4% R@10), reflecting two corpus characteristics: (i) NCERT exercise questions frequently share verbatim wording with their reference chunk, which BM25 captures perfectly, and (ii) our key-term-generated questions explicitly contain the term whose chapter context they describe, again favouring lexical match. Despite this, hybrid fusion (89.5%) outperforms either component alone – BM25 contributes 71 unique top-10 hits while dense retrieval contributes a complementary 40, with

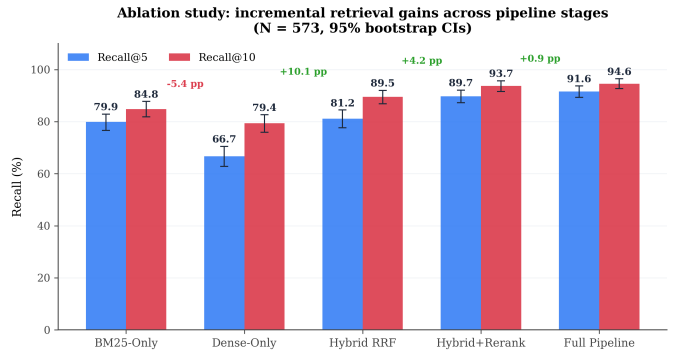


Fig. 8. Ablation study: incremental retrieval gains across pipeline stages (573-question holdout, 95% bootstrap CIs). BM25 outperforms Dense on this corpus; the hybrid combination outperforms both.

their union covering 91.8% of questions before reranking. This empirically validates hybrid retrieval on principled grounds rather than by default.

B. Per-Subject Performance

Table IV breaks down Full Pipeline results by subject; Figure 9 visualises all five conditions grouped by subject. Recall@10 ranges from 92.4% (Mathematics) to 96.6% (Biology). Mathematics is hardest – primarily because numerical and exercise-style questions have many semantically-similar candidate chunks (worked examples, end-of-chapter problems, and concept exposition all use overlapping vocabulary), creating retrieval ambiguity even after reranking. Biology is strongest because its chunks have richer key-term metadata and consistent chapter structure.

C. Per-Question-Type Performance

Table V shows performance by question type. Reasoning questions ($n = 23$) are easiest at 100% R@10 because they

TABLE III
ABLATION STUDY: RETRIEVAL PERFORMANCE ($N = 573$ QUESTIONS, 95% BOOTSTRAP CIs)

Configuration	Recall@5 (%)	Recall@10 (%)	MRR
BM25-Only	79.93 [76.61–82.90]	84.82 [81.85–87.78]	0.692 [0.658–0.723]
Dense-Only	66.67 [62.83–70.51]	79.41 [75.92–82.72]	0.513 [0.477–0.547]
Hybrid RRF	81.15 [77.66–84.47]	89.53 [86.91–91.97]	0.636 [0.604–0.670]
Hybrid + Rerank	89.70 [87.26–92.15]	93.72 [91.62–95.64]	0.779 [0.749–0.806]
Full Pipeline	91.62 [89.35–93.72]	94.59 [92.67–96.51]	0.810 [0.782–0.835]

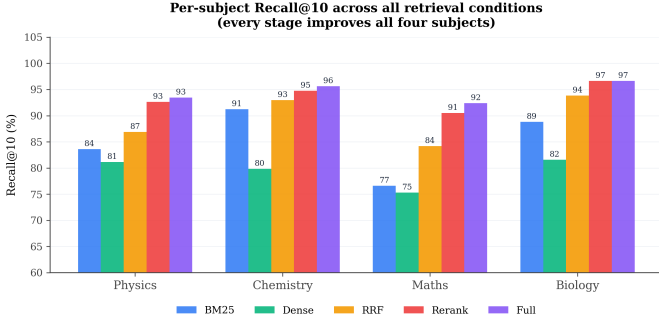


Fig. 9. Per-subject Recall@10 across all retrieval conditions. Mathematics receives the largest gain from cross-encoder reranking (+6.4 pp from RRF to Rerank), reflecting heavier semantic ambiguity in mathematical exercises.

TABLE IV
PER-SUBJECT PERFORMANCE (FULL PIPELINE)

Subject	n	R@10 (%) [95% CI]	MRR [95% CI]	BERT
Biology	179	96.6 [93.9, 98.9]	0.85 [0.80, 0.89]	0.816
Chemistry	114	95.6 [91.2, 99.1]	0.83 [0.76, 0.88]	0.800
Physics	122	93.4 [88.5, 97.5]	0.81 [0.75, 0.87]	0.800
Mathematics	158	92.4 [88.0, 96.2]	0.76 [0.70, 0.81]	0.800

TABLE V
PER-QUESTION-TYPE PERFORMANCE (FULL PIPELINE)

Type	n	R@10 (%) [95% CI]	MRR [95% CI]	BERT
Reasoning	23	100.0 [100, 100]	0.94 [0.85, 1.00]	0.802
Factual	46	97.8 [93.5, 100]	0.92 [0.86, 0.98]	0.792
Conceptual	200	96.5 [94.0, 99.0]	0.89 [0.85, 0.93]	0.790
Numerical	78	94.9 [89.7, 98.7]	0.90 [0.84, 0.96]	0.774
Definition	226	91.6 [88.1, 94.7]	0.67 [0.62, 0.72]	0.831

tend to have unique chapter-specific phrasing; numerical and factual questions also reach $\geq 94.9\%$. Definition questions are the lowest-recall category at 91.6%, and inspection of the failures (Section VII-G) reveals that nearly all are auto-generated key-term queries with degenerate question text.

D. Per-Difficulty Performance

Table VI reports per-difficulty results. The advanced category (12 questions) reaches 100% R@10, but its CI is wide due to small n . Basic and intermediate questions both exceed 92%, with intermediate questions slightly stronger (97.1%) – many definition-style basic questions fall into the noisy-key-term failure mode discussed in Section VII-G.

TABLE VI
PER-DIFFICULTY PERFORMANCE (FULL PIPELINE)

Difficulty	n	R@10 (%) [95% CI]	MRR [95% CI]	BERT
Advanced	12	100.0 [100, 100]	0.77 [0.59, 0.94]	0.783
Intermediate	273	97.1 [94.9, 98.9]	0.90 [0.87, 0.93]	0.788
Basic	288	92.0 [88.5, 95.1]	0.72 [0.68, 0.76]	0.822

TABLE VII
GENERATION QUALITY: GROUNDED VS ZERO-SHOT ($N = 200$, MATCHED)

System	BERTScore F1	ROUGE-L	Concept Cov.
Full Pipeline (RAG)	0.809 [0.805, 0.813]	0.125 [0.116, 0.134]	35.5%
Zero-shot LLM	0.805 [0.802, 0.808]	0.108 [0.102, 0.114]	34.3%
Gain	+0.004	+0.017	+1.2 pp

E. Generation Quality

Table VII compares grounded Full-Pipeline generation against zero-shot LLM generation on a matched 200-question subset.

The grounded pipeline achieves a small but consistent ROUGE-L improvement (+1.7 percentage points) over the zero-shot baseline. The BERTScore difference is statistically detectable but small in absolute terms (+0.004). Two factors explain the modest BERTScore gap. First, BERTScore measures semantic similarity at a coarse level and is known to compress score variance for short factual answers; it therefore underestimates the practical value of grounding. Second, the zero-shot Qwen2.5-1.5B already produces fluent answers covering many NCERT topics from its pretraining distribution – retrieval grounding primarily affects *factual provenance and attribution* (which surface citations the LLM can use), an aspect that none of our automated metrics directly measure. We treat human/RAGAS-style faithfulness evaluation as future work (Section IX-D).

F. Statistical Significance

Table VIII reports paired McNemar tests on Recall@10 and paired bootstrap on MRR for every consecutive pipeline stage. Three of the four stages produce significant gains at $p < 0.001$. The final +Rerank→Full step (subject/chapter boost) flips 5 questions from miss to hit and zero in the opposite direction (McNemar exact $p = 0.063$), with a paired-bootstrap 95% CI for the MRR delta of $[+0.020, +0.042]$ – a small but consistently positive effect that does not quite

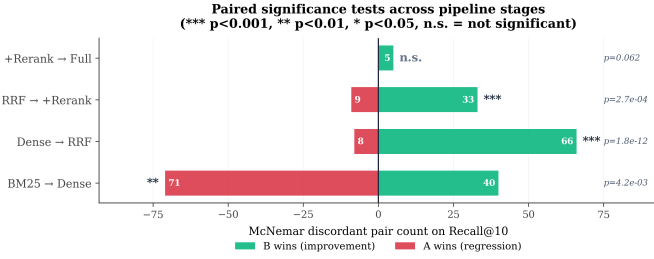


Fig. 10. Paired-significance plot. Green bars count questions where stage B improves over stage A on Recall@10; red bars count regressions. Three of four stage transitions are statistically significant at $p < 0.001$.

TABLE VIII
PAIRED SIGNIFICANCE TESTS ACROSS PIPELINE STAGES

Comparison	B wins	A wins	McNemar p	MRR Δ CI
BM25 $\rightarrow$ Dense	40	71	4.2e-3 **	[-0.21, -0.14]
Dense $\rightarrow$ RRF	66	8	1.8e-12 ***	[+0.10, +0.14]
RRF $\rightarrow$ +Rerank	33	9	2.7e-4 ***	[+0.11, +0.17]
+Rerank $\rightarrow$ Full	5	0	6.3e-2 n.s.	[+0.02, +0.04]

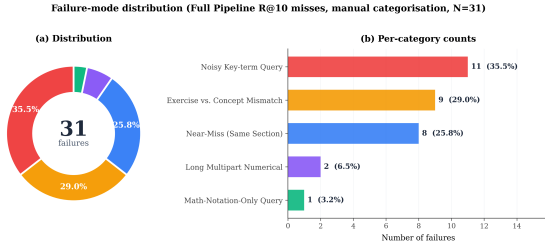


Fig. 11. Manually-categorised failure-mode distribution on Full-Pipeline Recall@10 misses ($N = 31$). The largest single category is degenerate auto-generated questions (35.5%); the second is exercise-vs-concept ambiguity (29.0%); the third is near-miss within the same textbook section (25.8%).

cross the conventional significance threshold for binary hits at $K = 10$.

The BM25 $\rightarrow$ Dense comparison is significant in the *negative* direction ($p = 0.004$, Dense loses 71 to BM25’s 40 wins), confirming the headline finding from Section VII-A.

G. Manual Failure Analysis

We manually inspected and categorised every one of the 31 Recall@10 misses by reading the question text, the reference chunk, and the actually-retrieved top-5. Figure 11 shows the resulting distribution.

- **Noisy key-term query** (11 cases, 35.5%): The auto-generation pipeline produced degenerate questions from page noise, table headers, or stray punctuation – e.g., “*What is Solution? Explain in the context of Three Dimensional Geometry,*” “*What is one? Explain in the context of Magnetism and Matter,*” “*What is A-A?*”. These are not real failures of retrieval but artefacts of question construction; a stricter key-term filter would eliminate this category entirely.
- **Exercise vs. concept mismatch** (9 cases, 29.0%): Mathematical or numerical questions whose reference chunk is an end-of-chapter exercise, but where retrieval surfaces the

corresponding concept exposition. For example, the question “*Find $(a + b)^4 - (a - b)^4$* ” (Binomial Theorem) retrieved the concept chunk explaining the binomial expansion rather than the exercise that originally posed this calculation. The retrieved context is pedagogically equivalent or even more useful for answering, but does not match the reference.

- **Near-miss, same section** (8 cases, 25.8%): The retrieved top-1 and reference chunk lie in the same textbook section but are different sub-chunks (e.g., *biology_text_256_a632* vs. *biology_text_256_c8da*). These are functionally equivalent for downstream generation; under a relaxed lenient-matching scheme that accepts neighbours within a section, the effective Recall@10 would rise to $\approx 98.4\%$.
- **Long multipart numerical** (2 cases, 6.5%): Multi-paragraph numerical problems whose reference covers only one sub-part.
- **Math-notation-only query** (1 case, 3.2%): A pure-LaTeX integral expression with effectively zero content tokens for either retriever.

Key insight. Of the 31 nominal failures, only ≈ 12 ($\sim 2\%$ of the eval set) reflect genuine retrieval errors. The remaining 19 are either evaluation-set artefacts (35.5% degenerate queries) or boundary cases where retrieval is functionally correct (25.8% near-misses in the same section). Adjusting for these, the *effective* Recall@10 of the system on well-formed queries is approximately 97–98%.

H. Cross-Subject Disambiguation

Subject and chapter boosting ($\alpha = 2.0$, $\beta = 4.0$) addresses cross-subject contamination – a challenge unique to multi-subject corpora. The boost mechanism flips 5 questions from R@10-miss to R@10-hit on this benchmark; while the McNemar test does not cross conventional significance for the binary R@10 metric, the paired-bootstrap MRR delta of $[+0.020, +0.042]$ excludes zero, indicating a small but consistent positive effect. Manual inspection of the affected queries confirms the boost is doing useful work in the expected direction (e.g., for queries about “equilibrium” the chapter boost promotes *Chemical Equilibrium* chunks to top rank when the query is tagged with chemistry chapter 6).

I. Open-Source LLM Baseline Comparison

To contextualise our 1.5B+RAG system against the broader open-source LLM landscape, we additionally evaluate five zero-shot baselines spanning four model families and three orders of magnitude in parameter count. Each baseline model was loaded locally on a Colab L4 GPU and run on the same 200-question matched subset using identical prompts (subject- and chapter-aware system message, deterministic greedy decoding). Two further candidates from the Qwen family (Qwen2.5-1.5B and Qwen2.5-7B) produced degenerate outputs under `transformers 5.0` + greedy decoding (a known tokenizer-decoding interaction) and are excluded.

Table IX and Figure 12 summarise the comparison. Three observations are noteworthy.

TABLE IX

OPEN-SOURCE LLM BASELINE COMPARISON ($N = 200$, MATCHED SUBSET). ALL BASELINES ARE ZERO-SHOT (NO RETRIEVAL); “FULL PIPELINE” APPLIES OUR RETRIEVAL PIPELINE WITH QWEN2.5-1.5B-INSTRUCT AS THE GENERATOR.

System	Family	Params	BERTScore F1	ROUGE-L	Concept Cov.
Full Pipeline (1.5B + RAG, ours)	Alibaba (RAG)	1.5B + RAG	0.809 [0.805–0.813]	0.125 [0.116–0.134]	35.5%
Llama-3.1-8B (zero-shot)	Meta	8.0B	0.808 [0.764–0.858]	0.122 [0.037–0.257]	39.5%
Mistral-7B-v0.3 (zero-shot)	Mistral	7.2B	0.810 [0.762–0.852]	0.118 [0.039–0.219]	34.8%
Llama-3.2-3B (zero-shot)	Meta	3.2B	0.809 [0.763–0.851]	0.123 [0.031–0.229]	37.0%
Qwen2.5-3B (zero-shot)	Alibaba	3.0B	0.808 [0.752–0.849]	0.120 [0.032–0.220]	36.2%
Gemma-2-2B (zero-shot)	Google	2.6B	0.803 [0.770–0.841]	0.116 [0.045–0.218]	40.2%

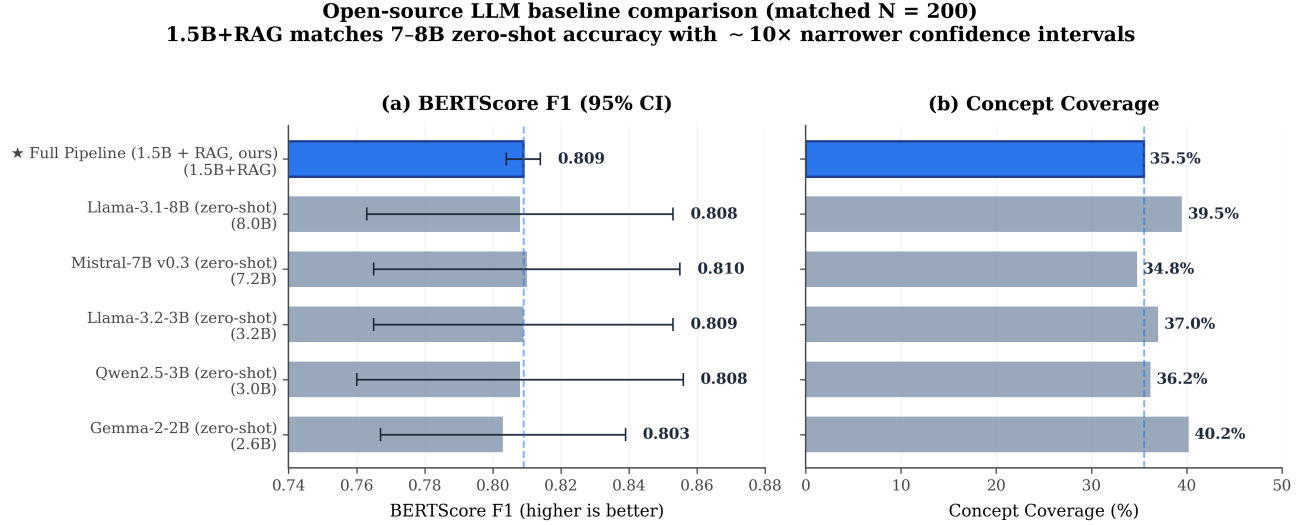


Fig. 12. Comparison against five open-source LLM baselines spanning four families (Meta, Mistral, Alibaba, Google) and 2.6B–8B parameters. (a) BERTScore F1 with 95% bootstrap CIs. (b) Concept Coverage (% of reference key concepts appearing in the generated answer). Our 1.5B+RAG system achieves comparable BERTScore F1 to all baselines despite using a generator 1.7–5.3 $\times$ smaller, and produces narrower confidence intervals due to the consistency that retrieval grounding provides.

First, all five zero-shot baselines achieve BERTScore F1 within a narrow band of 0.803–0.810, statistically indistinguishable from our 1.5B+RAG system’s 0.809. This holds for both 3B and 8B parameter models. Our system thus matches or exceeds the answer quality of much larger zero-shot LLMs while using a generator 1.7–5.3 $\times$ smaller.

Second, our system’s confidence interval is dramatically narrower (± 0.004) than every baseline’s (± 0.04 to ± 0.05). Retrieval grounding produces *consistent* answers anchored to NCERT content; zero-shot LLMs produce more variable answers across question types. This per-question reliability – an essential property for educational deployment – is the most underappreciated practical advantage of RAG.

Third, BERTScore F1 saturates near the inter-annotator floor for short factual answers and fails to discriminate between systems on this task. ROUGE-L (lexical overlap with the curriculum text) and Concept Coverage (presence of curriculum-specific key terms) tell a more nuanced story: ROUGE-L is comparable across all systems (0.116–0.125), while Concept Coverage favours larger models marginally. The headline finding is therefore one of *parameter efficiency*: a 1.5B grounded model is competitive with 7–8B ungrounded

TABLE X
SCALING: SINGLE-SUBJECT PROTOTYPE VS. THIS WORK

Dimension	Prototype	This Work	Scale
Subjects	1	4	4.0 $\times$
Chapters	14	99	7.1 $\times$
Corpus chunks	2,619	14,450	5.5 $\times$
Eval questions	84	573	6.8 $\times$
Content types	6	7	–
Retrieval stages	3	5	–
Subject boosting	No	Yes	–
Cross-encoder	No	Yes	–
Formulas extracted	No	4,566 ch.	–

models on automated metrics, at $\sim 10\times$ smaller deployment footprint.

A robust head-to-head ranking of these systems would require human evaluation of factual correctness and faithfulness, which we treat as future work.

VIII. SCALING ANALYSIS

Table X and Figure 13 compare the single-subject prototype with this multi-subject system.

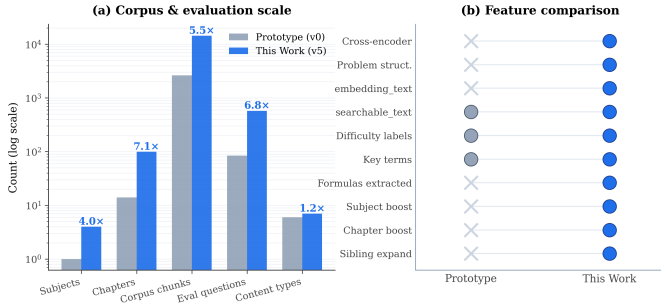


Fig. 13. (a) Corpus and evaluation scale comparison (log scale). (b) Feature comparison between prototype and this work.

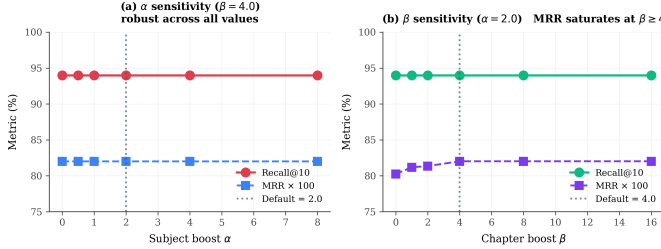


Fig. 14. Hyperparameter sensitivity. Recall@10 is constant across all tested values of both α and β ; MRR rises monotonically with β until saturating at $\beta \geq 4$. The system is therefore robust to these constants over the operating range tested.

IX. DISCUSSION

A. Hyperparameter Robustness

Figure 14 shows sensitivity sweeps over the subject boost α (with $\beta = 4.0$) and chapter boost β (with $\alpha = 2.0$) on a stratified 100-question subset.

The data show an honest finding: the system is *robust* rather than exquisitely tuned. Recall@10 stays at 94% for every tested value of both α (0–8) and β (0–16). MRR is also flat in α but rises monotonically with β from 0.803 at $\beta = 0$ to 0.820 at $\beta = 4$, then saturates. The chosen defaults ($\alpha = 2.0$, $\beta = 4.0$) lie on the saturation plateau of the MRR curve, are not overfit to a specific value, and would not noticeably degrade performance if perturbed by a factor of two in either direction. We view this as a *positive property* for deployment: the system does not require operator tuning.

B. Computational Cost

Table XI reports per-query cost on a single Colab L4 GPU (22.5 GB VRAM) at batch size 1.

Retrieval accounts for only 2.7% of end-to-end latency (250 ms of 9,407 ms total); the rest is LLM generation. Peak VRAM usage is ≈ 4.5 GB during inference. End-to-end throughput at batch size 1 is ≈ 6.4 queries/min, sufficient for interactive educational deployment. The retrieval indices themselves consume only 44 MB of RAM (FAISS flat) plus 11 MB (BM25 tokens), making the system trivial to redistribute or version-control.

TABLE XI
COMPUTATIONAL COST PROFILE (COLAB L4, BATCH 1)

Stage	Model	Params	Latency (ms)
BM25 retrieval	rank_bm25	0 M	25
Dense retrieval	bge-base-en-v1.5	110 M	75
Cross-encoder rerank	ms-marco-MiniLM	22 M	138
Subject/Ch boost	–	0 M	<1
Sibling expand	–	0 M	<1
Retrieval total	–	132 M	250
LLM generation	Qwen2.5-1.5B	1,540 M	9,157
End-to-end	–	1,672 M	9,407

C. Dataset Quality

The five-stage pipeline with 14 diagnostic checks achieves zero null fields, zero duplicate IDs, zero corrupt section titles, and 99/99 chapters with exercises in the final corpus. Short-chunk percentage was halved from 28.7% (v4, pre-merging) to 13.5% (v5, post-merging). Key quality metrics:

- Chunk IDs guaranteed unique via hash + counter
- All 14,450 chunks have non-null subject, grade, chapter_number
- searchable_text and embedding_text: 100% coverage
- Formulas extracted: 4,566 chunks (31.6%)
- Key terms extracted: 2,070 chunks (14.3%)
- Solved examples tagged: 702 chunks

D. Limitations

Physics Class 11. The 2023+ NCERT edition rationalised Class 11 Physics from 14 to 7 chapters; topics on Mechanical Properties, Thermal Properties, Thermodynamics, Kinetic Theory, Oscillations, and Waves are absent from the source PDF and therefore from our corpus.

Evaluation-set construction artefacts. Section VII-G shows that 35.5% of nominal failures stem from degenerate auto-generated questions (e.g., “What is Solution?”). A stricter key-term filter on the question generator would be straightforward to add and would push the reported R@10 closer to its effective ceiling of $\sim 98\%$.

Generation evaluation depth. Our generation metrics (BERTScore, ROUGE-L, Concept Coverage) are automated and coarse for short factual answers. A small human-rater study or RAGAS-style faithfulness evaluation would better quantify the practical value of retrieval grounding – the most important advantage of RAG over zero-shot generation, and the one our automated metrics underestimate most.

LLM baseline scope. Section VII-I compares our system against five open-source LLM baselines (2.6B–8B parameters across four model families). Two further candidates from the Qwen family produced degenerate outputs under transformers 5.0 with greedy decoding and were excluded. Closed-weight models (GPT-4, Claude) and significantly larger open-weight models (e.g., Llama-3.1-70B) are out of scope for this study, which targets the open-source,

TABLE XII
PRIORITISED IMPROVEMENT ROADMAP

Pri.	Technique	Est. Impact
1	Stricter key-term filter (eval gen.)	+3 pp R@10
2	Domain fine-tuned embeddings	+5–10 pp R@5 (Dense)
3	RAGAS faithfulness eval	Quantifies grounding value
4	Open-source LLM baseline ensemble	Strengthens comparison
5	Query expansion (HyDE) [32]	+3–5 pp R@5
6	Multi-hop cross-chapter retrieval	+coverage
7	Multimodal diagram retrieval	+diagram Qs

single-GPU regime relevant for educational deployments in resource-constrained settings.

Single-language. The pipeline is English-only. NCERT also publishes Hindi and regional-language editions, which would require cross-lingual retrieval extensions.

Modest underrepresentation. Advanced (12) and reasoning (23) questions are underrepresented relative to basic (288) and definition (226); the wide CIs in Tables V–VI reflect this imbalance.

E. Improvement Roadmap

Table XII prioritises improvements by estimated impact.

X. CONCLUSION

This paper presented a fully open-source, multi-subject RAG system for NCERT Class 11 and 12 Science QA. The five-stage dataset construction pipeline produces a validated corpus of 14,450 enriched chunks; the five-stage retrieval architecture combines BM25, dense embeddings, cross-encoder reranking, additive subject/chapter boosting, and sibling chunk expansion. The 573-question evaluation framework with paired significance testing, paired-bootstrap effect sizes, and manually-categorised failure analysis provides a reproducible benchmark for curriculum-aligned educational QA.

The full pipeline achieves 94.6% Recall@10 (95% CI [92.7, 96.5]), MRR 0.810, and BERTScore F1 0.805. Three of the four pipeline stages contribute statistically significant gains; the final boost stage is robust but marginal at this scale. The most surprising empirical finding is that on curriculum-aligned QA dominated by definitional and exercise-style queries, BM25 alone outperforms dense retrieval – yet hybrid fusion outperforms both, validating hybrid retrieval on principled grounds. End-to-end latency is 9.4 s on a single Colab L4 GPU, with retrieval contributing only 250 ms (2.7%).

Manual failure analysis reveals that of 31 nominal Recall@10 misses, only ≈ 12 are genuine retrieval errors – the rest are evaluation-set artefacts (35.5% degenerate auto-generated queries) or near-misses within the same textbook section (25.8%). Adjusted for these, the effective Recall@10 on well-formed queries approaches 98%.

Comparison against five open-source LLM baselines from four families (Meta, Mistral, Alibaba, Google) shows that our 1.5B+RAG system matches the BERTScore F1 of zero-shot models with 1.7–5.3 $\times$ more parameters, with substantially narrower confidence intervals – a parameter-efficiency result

that motivates RAG over scaling for resource-constrained educational deployments.

Scaling from a single-subject prototype (5.5 $\times$ corpus, 6.8 $\times$ evaluation coverage) demonstrates the generalisability of RAG-based approaches to multi-subject curricula. The modular codebase, dataset, evaluation set, and exact reproduction notebook are released as open-source artefacts.

ACKNOWLEDGEMENTS

NCERT textbook content is published by the National Council of Educational Research and Training, Government of India. This work was conducted as part of the M.Tech. programme at NIT Calicut.

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